

Article

Research on Dual-Emotion Feature Fusion and Performance Improvement in Rumor Detection

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Abstract: At present, a large number of rumors are mixed in with various kinds of news, such as current affairs, politics, social economy, and military activities, which seriously reduces the credibility of Internet information and hinders the positive development of various fields. In previous research on rumors, most scholars have focused their attention on the textual features, contextual semantic features, or single-emotion features of rumors but have not paid attention to the chain reaction caused by the hidden emotions in comments in social groups. Therefore, this paper comprehensively uses the emotional signals in rumor texts and comments to extract emotional features and determines the relationship between them to establish dual-emotion features. The main research achievements include the following aspects: (1) this study verifies that, in the field of affective characteristics, the combination of rumor-text emotion and comment emotion is superior to other baseline affective characteristics, and the detection performance of each component is outstanding; (2) the results prove that the combination of dual-emotion features and a semantic-feature-based detector (BiGRU and CNN) can improve the effectiveness of the detector; (3) this paper proposes reconstructing the dataset according to time series to verify the generalization ability of dual affective features; (4) the attention mechanism is used to combine domain features and semantic features to extract more fine-grained features. A large number of data experiments show that the dual-emotion features can be effectively compatible with an existing rumor detector, enhance the detector's performance, and improve the detection accuracy.



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Keywords: dual-emotion features; BiGRU; CNN; domain embedding; F1 score

1. Introduction

At present, with the increasing level of Internet technology, social self-media has flooded people's lives and work, which also encourages everyone to realize freedom of speech [1]. As particularly important components of social media, microblogs, Toutiao, public accounts, and TikTok not only provide information and social platforms for network users but also set up network hotbeds for the spread of confusing social rumors. The normalization of network media has a profound impact on people's living conditions, daily travel, and other aspects. However, it also causes a lot of harm, because the network amplifies the characteristics of information concealment and allows different users to publish their opinions anytime and anywhere [2]. Therefore, the phenomenon of false information and even rumors flooding the entire social network has become more and more difficult to control. Rumors manipulate public perception to a great extent, and their negative manifestations are mainly weakened public trust in government, damage to the reputations of well-known people running election campaigns and other public figures, and economic effects on the consumption of products and food.

Scholars have focused on rumor detection using various methods, from artificial rumor detection methods at the beginning to detection methods based on machine learning

and then to detection methods based on deep learning [3]. Their characteristics are as follows: manual detection can guarantee accuracy, but deficiencies such as the delay in detection results and the inability to efficiently process huge rumor datasets have always existed [4]. Machine learning-based detection makes up for the deficiencies of manual detection in time and effort, but the feature extraction generalization ability is poor, which is not conducive to the detection of multi-class targets, and the detection accuracy is limited. Deep learning is a method that simulates the thinking mode of the human brain through program coding, trains computers to automatically extract text features, and combines feature learning with model building to make up for the incompleteness of artificially designed feature extractors [5]. According to a survey, researchers have generally recognized detection methods based on deep learning, but in practice, they are still attacked in multiple dimensions. In view of the complexity and variability of the category, generation time, publishing language, and acquisition platform of rumors, scholars have conducted a variety of validation studies on training models and explored research directions for this task, but which method has the best effect is still unclear [6].

Due to the lack of a systematic understanding of economic, political, military, social, and other matters, the thoughts and emotions of social groups are easily incited by fierce language, pictures, and videos. Therefore, if rumors are quickly detected in the process of spreading and official media promptly issues a statement to refute them, the social harm of the rumors can be minimized. Research on rumor detection will play an important role in controlling and mitigating the governance of the social network environment, the control of social public opinion, and conflicts among countries [7]. In this paper, emotional signals in rumors and comments are used to extract emotional features, and the relationship between them is determined to establish dual-emotion features. The main research achievements include the following aspects:

- (1) **Rumor-Text Emotion and Comment Emotion:** These two elements form the basis of the dual-emotion feature approach. Rumor-text emotion refers to the emotional sentiment conveyed by the rumor text itself, while comment emotion captures the emotional reactions and sentiments expressed in comments on the rumor. By combining these two emotional signals, a more comprehensive understanding of the rumor's emotional impact can be achieved. This combination of emotions is found to be more effective than focusing on either textual, contextual, or single-emotion features alone.
- (2) **Dual-Emotion Features and Semantic-Feature-based Detector:** Dual-emotion features are used in conjunction with a semantic-feature-based detector, such as a BiGRU (Bidirectional Gated Recurrent Unit) and CNN (Convolutional Neural Network). The semantic features extracted from the text help in understanding the context and meaning of the rumor, while the dual-emotion features add the emotional dimension. This combined approach enhances the effectiveness of the detector by providing a richer, more nuanced view of the rumor.
- (3) **Reconstructing the Dataset According to Time Series:** This step is used to validate the generalization ability of dual affective features. By reconstructing the dataset based on time series, researchers can test how well the dual-emotion features perform when applied to rumors over different time periods. This helps ensure that the approach is not overly specific to a particular set of rumors or a narrow time frame but can be generalized to a wider range of rumors.
- (4) **Attention Mechanism for Combining Domain Features and Semantic Features:** The attention mechanism is used to effectively combine domain-specific features (which may include knowledge about the subject matter of the rumor) with semantic and dual-emotion features. This allows the model to focus on the most relevant and informative aspects of the data, extracting more fine-grained features that further improve the performance of the rumor detector.

A large number of data experiments show that dual-emotion features can be effectively compatible with an existing rumor detector, enhance the detector's performance, and improve the detection accuracy.

Section 2 is a literature review of recent work, introducing the status of rumor detection and the application of feature fusion methods in rumor detection. Section 3 introduces the proposed model and its different components. In Section 4, the datasets, experimental settings, and experimental results are introduced in detail, and the ablation results are fully discussed. Section 5 gives the conclusions according to the research presented in this paper. The links between the parts of the article are shown in Figure 1.

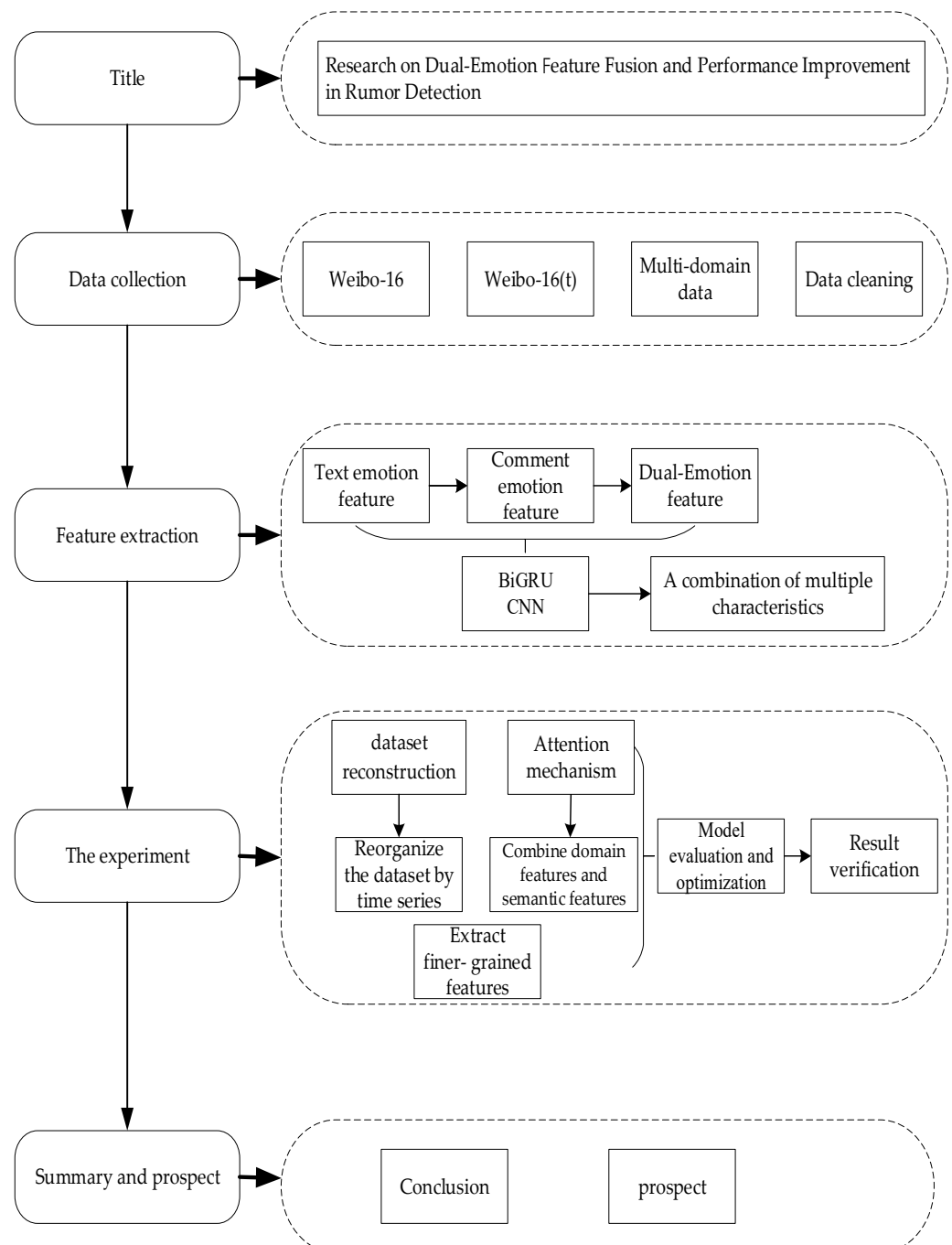


Figure 1. The links between the parts of the article.

2. Related Work

In recent years, the field of rumor detection has attracted the attention of scholars and researchers from computer science, information science, sociology, psychology, and other disciplines. They are committed to developing efficient and accurate rumor detection

algorithms and techniques to cope with the increasingly complex rumor propagation situation. Next, we provide a detailed overview of the relevant work in this field, including the latest research results, technical methods, challenges, and future directions.

2.1. Starting with Model Selection

In recent years, scholars have been exploring rumor detection models. One of the first machine learning-based models to emerge was one that learned to recognize rumors from training data. Such models can automatically learn new rumor forms and detect them without human intervention. Their advantages are strong adaptability and good scalability, but the disadvantage is that a large amount of data are required to train the model, and the accuracy may be affected by the quality of the training data. Such models include naive Bayes classifiers, decision trees, support vector machines (SVMs), and so on. With the development of technology, multimodal rumor detection models have attracted more and more attention. This kind of model not only considers text information but also integrates multiple modes of information, such as pictures, videos, and user behaviors, to detect rumors. Wang Yuyan et al. proposed joint detection using a Tanh-RNN model, LSTM, and a GRU model and made two datasets for evaluation [8]. Yu et al. proposed a method for identifying error information based on a CNN model (CAMI) [9]. Guo et al. proposed a BiLSTM model that combined the hierarchical network framework with manually established social features [10]. Nguyen et al. combined CNN and RNN models, using LSTM to represent tweets, and provided training sets from online rumor collection websites, with an accuracy rate of 81.9% [11]. Jin et al. adopted the RNN + attention mechanism model and integrated multimodal features, cross-combined the features of texts, pictures, and videos, and tested the method on datasets from Twitter and Weibo, respectively, with the highest probabilities of 77.8% and 68.2% [12]. Alkhodair et al. combined an LSTM-RNN model with a Word2Vec model to compare the evaluation data before and after adding grammatical features and social features, but the accuracy was not significantly improved [13]. Asghar et al. used the embedded layer of a BiLSTM and CNN model for testing and evaluation, and the result was 86.12% [14]. Lin et al. proposed a zero-shot response-aware prompt learning framework to bridge language and domain gaps in rumor detection. They presented a prompt-based approach to avoid reliance on language-specific rumor prompt engineering, with effective response fusion strategies to incorporate influential and structural propagation threads for domain adaptation [15]. For Zhang et al., their intuition was to exploit diverse counterfactual evidence in an event graph to serve as multi-view interpretations, which were further aggregated for robust rumor detection results [16]. The inconsistencies between modalities are vital, so Yang et al. developed a novel deep visual–linguistic fusion network (DVLFN) that considers cross-modal inconsistencies and detects rumors by comprehensively considering modal aggregation and contrast information.

The evolution of rumor detection models from traditional machine learning approaches like naive Bayes, decision trees, and SVMs to more sophisticated multimodal frameworks highlights the increasing complexity and sophistication of the field. The integration of multiple data modalities, including text, images, videos, and user behaviors, has significantly improved the accuracy and robustness of rumor detection systems. The studies mentioned above, such as Wang Yuyan’s joint detection model, Yu’s CNN-based method, Guo’s BiLSTM model, and Jin’s RNN+Attention model, demonstrate the diverse strategies researchers have employed to tackle the challenge of rumor detection.

However, despite these advancements, there remains a need for models that can effectively handle the intermodal inconsistencies and domain gaps that arise when fusing information from different sources. The works by Lin et al. on zero-shot response-aware prompt learning and Yang et al. proposes DVLFN model which represent promising directions in this regard, aiming to bridge the gaps between languages and domains while considering cross-modal inconsistencies [17].

In this context, our work aims to build upon these foundations by proposing a novel multimodal rumor detection model that not only integrates diverse data modalities but also addresses the challenges of intermodal inconsistencies and domain adaptation in a more comprehensive manner. By leveraging advanced deep learning techniques and innovative fusion strategies, we aim to develop a model that can achieve even higher accuracy and robustness in detecting rumors across different platforms and contexts. Common rumor detection models are shown in Figure 2.

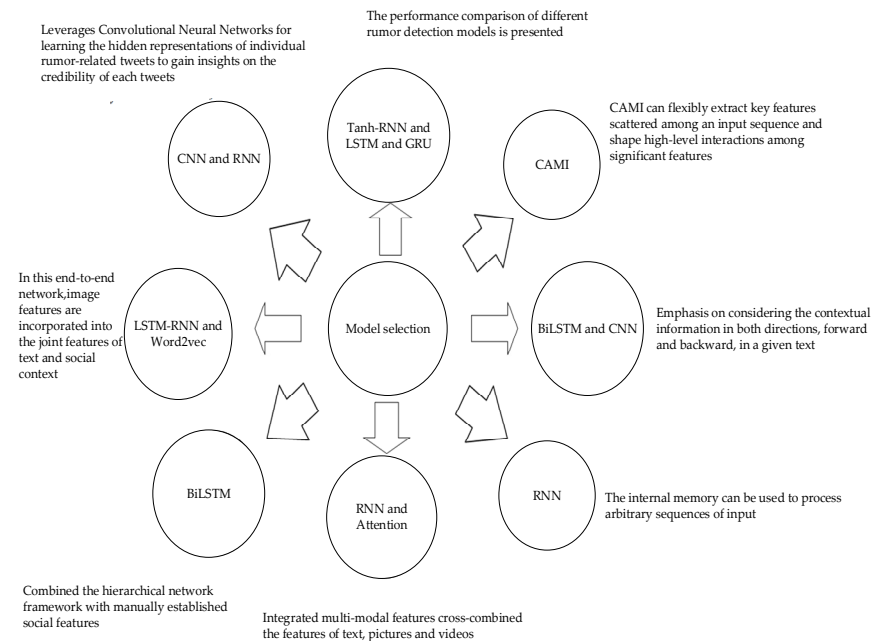


Figure 2. Common rumor detection models.

2.2. Starting with Feature Analysis

Chen et al. [18] proposed a novel graph convolution network model named the multilevel feature fusion-based graph convolution network (MFF-GCN), which employs multiple streams of GCNs that each learn different-level features of rumor data. In [19], a multi-feature fusion neural network with an attention mechanism was proposed for rumor detection, which makes an attempt to integrate user, textual, and propagation features into one united framework. Specifically, a Bidirectional Long Short-Term Memory Network (Bi-LSTM) is applied to extract user and textual features, and a Graph Convolutional Neural Network (GCN) is employed to extract high-order propagation features.

Ajao et al. verified the relationship between the authenticity of news (rumor or non-rumor) and the use of sentimental words and designed the emotional features (the ratio of the number of negative to positive words) [20]. Giachanou et al. did not add a single feature but extracted richer emotional features from news content for rumor detection based on emotional vocabulary [21]. Zubiaga et al. proposed a new method that used a conditional random field (CRF) to learn from the sequence dynamics of social media posts by using content features and social features [22]. The results showed that the F1 score was 60.7%, with low accuracy. Ghanem et al. explored the impact of emotional characteristics on the detection of different types of misinformation (i.e., propaganda, hoax, clickbait, and sarcasm) [23]. In 2020, Wu and Rao proposed an adaptive fusion network for rumor detection that models affective embeddings from content and comments, but it does not explore the specific differences in dual affective signals between rumors and non-rumors [24]. Zhang et al. [25] paid attention to the fact that the emotions (i.e., social emotions) aroused by news comments in the population cannot be ignored and verified the difference between dual emotions in fake news and real news. Haque et al. [26] extracted emotional and psycholinguistic features from posts and comments to enhance their ap-

proach and make rumor detection more effective. Al-saif et al. [27] used recent advances in natural language processing (NLP) to introduce a context-aware rumor detection method for Arabic social media. It can be seen that many experiments by scholars have proved that methods for rumor detection still have infinite possibilities, and how to give full play to the role of emotional signals needs to be further studied and explored.

The current research on emotion analysis is developing in a more in-depth and comprehensive direction by not only focusing on emotion words and sentimental words in the text but also combining content features and social features, using big data and machine learning technology to build emotional models, and exploring the relationship between rumors and emotional features. These studies not only improve the accuracy and efficiency of sentiment analysis but also provide powerful analytical tools for psychology, sociology, marketing, and other fields. The analysis of emotional characteristics is shown in Figure 3.

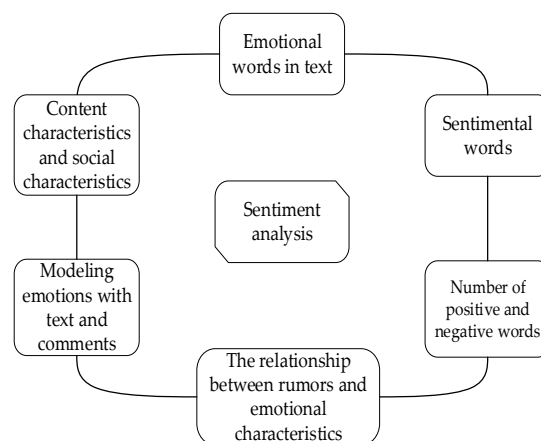


Figure 3. Emotional feature analysis diagram.

In order to visually show the rumor detection methods mentioned in different references, this paper compares the characteristics of different methods in Table 1.

Table 1. A summary of existing detection approaches for fake news.

Ref	Year	Dataset	Classification Approach	News Content	Social Context	Emotion Feature	Results
[20]	2019	PHEME labeled Twitter dataset	Different classification approaches	✓		✓	The use of emotional words is beneficial in sentiment-aware misinformation detection
[21]	2019	Politifact-1 Politifact-2	EmoCred	✓		✓	Outperforms LSTM-text by 12.39% in terms of F1 score
[22]	2017	Twitter (five events)	Conditional random fields	✓	✓		F1 score of 0.607
[23]	2020	News Articles and Twitter	Emotionally Infused Network	✓		✓	Superior results with F1 value of nearly 96%
[24]	2020	RumourEval PHEME	AIFN	✓	✓		Boosts accuracy by more than 2.05% and 1.90%
[25]	2021	RumourEval-19 Weibo-16 Weibo-20	BiGRU		✓	✓	Outperforms state-of-the-art task-related emotional features
[26]	2023	PHEME	Emotion-Enriched and Psycholinguistic Feature-Based Approach		✓	✓	Superior performance

3. Methodology

Before discussing the experimental design and implementation of rumor detection in depth, it is necessary to clarify the importance and purpose of the experiment. As a key information processing technology, the rumor detection experiment is not only a key link in verifying the effectiveness of theoretical hypotheses and algorithms but also an important driving force to promote technological progress and applications.

Through well-designed experiments, we were able to systematically evaluate the performance of different rumor detection methods, including key metrics such as accuracy, recall, and F1 scores. These metrics not only help us understand how well the algorithm performs on a particular dataset but also reveal the applicability and limitations of the algorithm in different scenarios.

In addition, the experimental part also undertakes the important task of exploring and optimizing the algorithm. In the process of experimentation, we can try different parameter settings, feature selections, model architectures, etc., to find the best algorithm configuration and further improve the accuracy and efficiency of rumor detection. At the same time, the diversity and complexity of the experimental data also provide us with rich scenarios to test the robustness and generalization ability of the algorithm.

Therefore, in the following content, we will introduce the experimental design, dataset selection, evaluation indicators, experimental process, and preliminary analysis of the experimental results of this rumor detection experiment in detail. Through this content, we hope to present a comprehensive, in-depth, and rigorous rumor detection experimental process for readers and provide a valuable reference for subsequent research and applications.

After selecting information datasets with the appropriate size and rich categories, this paper mines the emotional features in news texts and comments (i.e., emotion category, emotion intensity, emotion score, emotion vocabulary, and other auxiliary features). In order to make the following description more clear, this paper defines the features of publisher and social emotions (dual emotions). Then, the relationship and differences between dual-emotion features are analyzed, and all kinds of emotional feature sets are connected to form dual-emotion features so as to fully express dual emotions and the relationship between the two emotions. Finally, the data are inserted into the model trainer to compare the effectiveness and accuracy of each feature parameter in rumor detection using the same trainer and determine whether adding dual-emotion features can improve the accuracy of different model trainers. This paper mainly studies the following four issues:

(1) Extracting the dual-emotion features of rumors

Previous studies have all been based on textual features, semantic features, or a single emotional feature of texts, but few people have paid attention to the emotions triggered by comments on rumors, and the emotions conveyed in comments usually represent people's attitudes toward the content of a rumor. Therefore, this paper extracts the dual-emotion features of comment and rumor texts, outputs a feature vector, and sends the vector in series to the MLP layer and softmax layer for the final prediction of rumor accuracy.

(2) Determining whether dual-emotion features are more effective than baseline characteristics

In order to prove the necessity of dual-emotion features and the importance of exploring dual-emotion features and their interrelationships for rumor detection, this paper selected two baseline affective features to evaluate the validity of dual-emotion features. These features all use the same sentiment dictionary and are trained in the same model.

(3) Determining whether dual-emotion features enhance the performance of the detector

The characteristic components of dual emotions are publisher emotions, social emotions, and the emotion gap. An ablation study was used to determine whether they all enhance the detector's performance. To determine whether the components of dual-emotion features play a role on their own, we need to separately add the three emotional features to the model and compare the performance of dual-emotion features by conducting

experiments on different datasets. This also demonstrates the respective performance of publisher emotions, social emotions, and the emotion gap and whether the integration of the three exerts the maximum effect.

(4) Determining whether dual-emotion features can correctly process special rumor data

Rumor texts and comments on the Internet are highly unpredictable, and detectors trained in advance cannot immediately detect updated emotional information. Therefore, this paper studies whether dual-emotion features can accurately detect difficult rumor data.

3.1. Model Architecture

The experimental steps are as follows and the frame diagram of the model is shown in Figure 4:

- (1) Firstly, the emotion classification model and emotion dictionary are selected to identify emotion words that can represent the characteristics of rumors and comments to the greatest extent. Secondly, the Numberbatch word vector is applied; the emotional feature information is combined in the Numberbatch word vector, and the words representing the emotional features of rumors and texts are transformed into the form of an emotion vector, which is convenient for the computer system to recognize and use.
- (2) Different from rumor texts, comments have a large amount of data, and the emotion vector needs to transform each comment feature. To avoid a vector dimension disaster, mean pooling is used to calculate the average emotional signal of N comments, max pooling is used to obtain the extreme emotional signal of N comments, and then the comment emotion vector of two dimensions is output, and Concat is used to concatenate the two vectors. Finally, it is used to represent the emotional characteristics and social-emotional characteristics of the publisher.
- (3) By comparing the values between the two emotional characteristics, two phenomena can be obtained: the first is emotional resonance, where the emotional and social-emotional characteristics of the publisher are the same or similar; the second is emotional disharmony, where the emotional characteristics between the two are inconsistent. When the emotional features are inconsistent, the differences between the publisher's emotional features and the two feature dimensions of social emotions are used to obtain two dimensions of the emotion gap. The dual-emotion features are composed of the publisher's emotional feature, social-emotional features, and emotion gap.
- (4) The semantic embedded word vector is used to represent the lexical features of the rumors, and the semantic vector is input twice into the BiGRU model and the CNN model for semantic feature extraction.
- (5) The domain features of rumors are extracted and analyzed, and semantic features are combined with domain features through the attention mechanism to extract finer-grained features.
- (6) The dual-emotion features are spliced with BiGRU model features, CNN model features, and domain features, then input into the MLP, and finally classified through softmax to output the results.

3.2. BiGRU Model Algorithm

The GRU is an improvement of the recurrent neural network (RNN), which solves the problems of gradient explosion and gradient disappearance [28]. Building a BiGRU semantic analysis model requires the input of historical data into both forward GRUs and reverse GRUs to obtain maximum contextual information. The GRU has no cell state and uses gates to regulate the flow of information inside the unit, improving on the LSTM; the gates are reduced to two, namely, the update gate and the reset gate (z_t and r_t). BiGRU network structure diagram is shown in Figure 5.

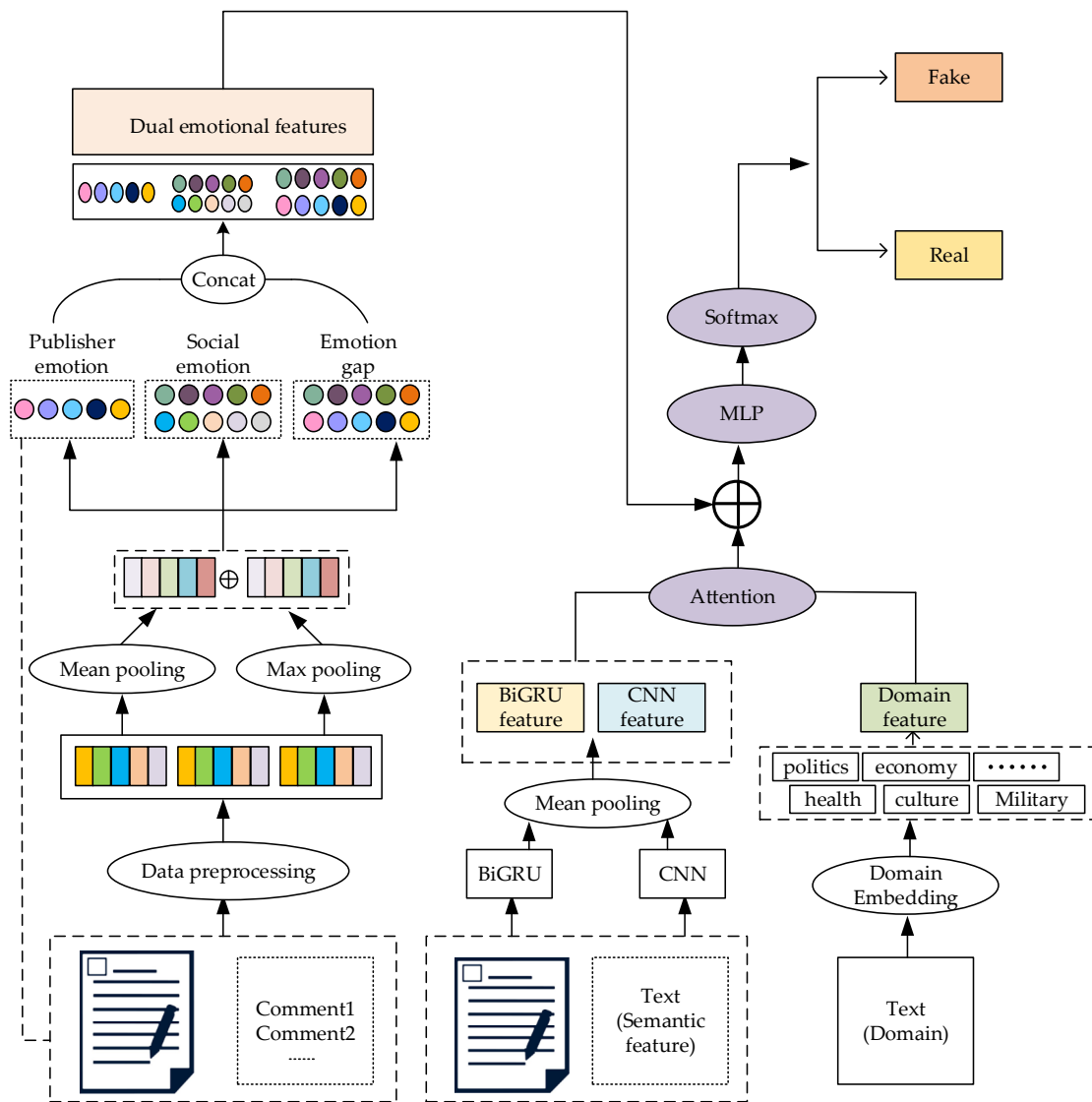


Figure 4. Overall architecture diagram of the model.

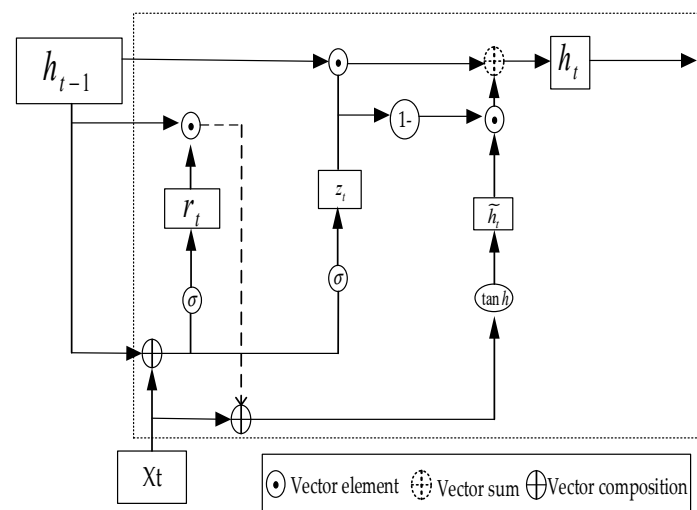


Figure 5. BiGRU network structure diagram.

The reset gate indicates the degree of information ignored at the previous time step. A smaller value indicates that more information is ignored.

$$r_t = \sigma(w_r x_t + U_r h_{t-1} + b_r) \quad (1)$$

The update gate controls how much information is retained from the previous moment to the current state and how much information is updated from the candidate state. The higher the value, the more data it remembers.

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (2)$$

$$h_t = (1 - z_t) \tilde{h}_t + z_t * h_{t-1} \quad (3)$$

$$\tilde{h}_t = \tanh(W_{\tilde{h}} x_t + U_{\tilde{h}}(h_{t-1}, r_t) + b_{\tilde{h}}) \quad (4)$$

In the model, x represents the input, h_t is the current hidden state, and $\tilde{h}_t$ is the previous hidden state. W is the weight matrix, b is the bias matrix, σ and $\tanh$ are activation functions, and U represents the randomly initialized attention matrix.

3.3. Convolutional Neural Network Model (CNN)

A CNN is an unsupervised multi-layer feedforward neural network, similar to a multi-layer perceptron (MLP). In addition to the input and output layers, the hidden layer of the CNN also includes other layers, namely, the convolutional layer, the pooling layer, and the fully connected layer (see Figure 6) [29].

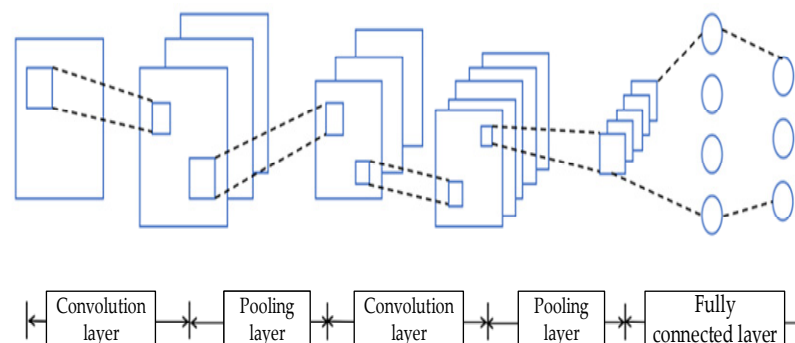


Figure 6. CNN structure diagram.

The convolution layer is connected to the input layer to set the size of a specific window (filter). Each filter has a one-dimensional convolution window length, and the filter (or feature detector) slides through the rumor to extract features of the lexical context. For example, a filter with a window size of 4 is used to deal with rumors that “social media has become the main source of information”. The convolutional layer extracts the features of “social media”, “media into”, and “media into” in turn and then transmits these feature elements to the next layer.

Pooling layer. The convolutional layer is equivalent to the input layer of the pooling layer and presents a many-to-one relationship. The main purpose of the pooling layer is to extract the most important features of each filter map to reduce the dimension of the features of the convolutional layer and reduce the computational burden of the CNN. The most widely used pooling methods are max pooling and mean pooling.

Fully connected layer. Each node of the feature dimension optimized by the convolution layer and pooling layer is connected to each node of the fully connected layer. The value is then passed to the output layer.

3.4. Extracting the Emotional Features of Text and Comments

After data preprocessing, a large number of redundant words in the rumors are removed, and then feature extraction is carried out on the cleaned data. Suppose a rumor consists of M sentences, where each sentence T consists of m words, and T^i is the i -th sentence in rumor M , i.e., $T^i = [x_1^i, x_2^i, \dots, x_m^i]$. A user comment corresponding to a rumor consists of N sentences, where each sentence L consists of n words, i.e., $L^j = [x_1^j, x_2^j, \dots, x_n^j]$, and $x_1x_2x_3 \dots x_n$ is represented by an embedded vector.

3.4.1. Extraction of Emotional Features of Publishers

In order to fully express the emotions of the publisher, various emotional features are extracted from the rumor text, including the emotion category, emotion vocabulary, emotion intensity, emotion score, and other auxiliary features. Of the five characteristics, the emotion category, emotion intensity, and emotion score basically define the overall emotional information of the rumor, and the other two provide information at the lexical and symbolic levels.

- (1) Emotion Category: An emotion classifier can effectively help identify emotion types in a more detailed and accurate way. Here, the Baidu AI model is used to directly output category features in rumor T .
- (2) Emotion Lexicon: High-frequency emotion words can express the specific emotions of rumors and are realized by using an emotion dictionary annotated by experts. For rumor T , t_i is the i -th word in T , and there are d_e emotional categories in the emotion dictionary, $E = \{E_1, E_2, \dots, E_{d_e}\}$.

Consider the frequency of t_i in the emotion dictionary and the semantic influence of the above and below words, such as degree words and negative words, which have the effect of changing the direction of the emotion (e.g., “unhappy”, “very fond”). Secondly, the degree value of negative words and degree words in the emotion dictionary and the probability of matching emotion words are used to calculate

$$s(t_i, e) = neg(t_i, w) * deg(t_i, w) / L \quad (5)$$

The value $s(T, e)$ of all words t on a particular emotion e is calculated, and the scores for each word in the range of emotion e are added:

$$s(t, e) = \sum_{i=1}^L s(t_i, e), \forall e \in E \quad (6)$$

W represents the window size, and $neg(t_j)$ and $deg(t_j)$ represent the negative word degree value and degree word dimension value, respectively:

$$neg(t_{i,w}) \prod_{j=i-w}^{i-1} neg(t_j) \quad (7)$$

$$deg(t_{i,w}) = \prod_{j=i-1}^{i-1} deg(t_j) \quad (8)$$

The scores of all the emotional texts are connected:

$$emo \frac{lex}{T} = s(T, e_1) \oplus s(T, e_2) \oplus \dots \oplus s(T, e_{d_e}) \quad (9)$$

- (3) Emotion Category: The process of extracting lexical intensity features is similar to that of extracting emotion vocabulary features. Intensity scores are involved here. For example, the intensity value of ecstasy is higher than that of happiness. First, the perceived intensity-level score of the text is calculated:

$$S'(T, e) = \sum_{i=1}^L s'(t_{i,e}) = \sum_{i=1}^L int(t_i) * s(t_{i,e}), \forall e \in E \quad (10)$$

$int(t_i)$ is the emotion intensity score of word t_i in the dictionary; if t_i is not in the dictionary, then $int(t_i) = 0$. The final emotion intensity feature is obtained by connecting the intensity perception scores:

$$emo_T^{int} = s'(T, e_1) \oplus s'(T, e_2) \oplus \dots \oplus s'(T, e_{d_e}) \quad (11)$$

- (4) Sentiment Score: Typically, a sentiment score is a positive or negative value that represents the degree of positive and negative polarity of the entire text, and it can be calculated by using a sentiment dictionary or a public toolkit.
- (5) Other Auxiliary Features: Considering that the above features do not explicitly utilize information outside the affective dictionary, we introduce a set of auxiliary features to capture the affective signals of non-verbal elements, including emoticons, punctuation, sentimental words, and personal pronoun frequencies. For example, emojis and punctuation marks are universal symbols for emotional expression around the world, such as “:)” and “!” Such punctuation marks can convey people’s emotions. The above five aspects of feature results are used to connect and extract the emotional features of publishers:

$$emo_T = emo_T^{cate} \oplus emo_T^{lex} \oplus emo_T^{int} \oplus emo_T^{senti} \oplus emo_T^{aux} \quad (12)$$

3.4.2. Extraction of Social-Emotional Characteristics

Given a rumor, the overall comment is represented by $M = \{M_1, M_2, \dots, M_i, \dots, M_{IM}\}$, where M_i represents comment i in M , and emo_{M_i} is obtained according to formula (13) in a similar way to the extraction of social-emotional features. Then, the transposed emotion vector of each comment is superimposed to obtain the emotion vector of all comments:

$$\widehat{emo} = emo_{M_1}^T \oplus emo_{M_2}^T \oplus \dots \oplus emo_{M_L}^T \quad (13)$$

After all emotion vectors are obtained, the social emotion of the whole comment list is generated by two aggregators: Formula (14) represents the average pool of average emotional signals; Formula (15) is used to capture the maximum pool of extreme emotional signals.

$$emo_M^{mean} = mean(\widehat{emo}_M) \quad (14)$$

$$emo_M^{max} = max(\widehat{emo}_M) \quad (15)$$

And finally, the above components are connected to form social-emotional traits:

$$emo_M = emo_M^{mean} \oplus emo_M^{max} \quad (16)$$

3.4.3. Extraction of the Emotion Gap

In order to study whether the relationship between dual-emotion features is helpful for rumor detection, the emotion gap is proposed, that is, the difference between publisher emotions and social emotions:

$$emo^{gap} = (emo_T - emo_M^{mean}) \oplus (emo_T - emo_M^{max}) \quad (17)$$

3.4.4. Generation of Dual-Emotion Features

The publisher’s emotional characteristics, social-emotional characteristics, and emotional characteristic gap are connected to form dual-emotion features:

$$emo^{dual} = emo_T \oplus emo_M \oplus emo^{gap} \quad (18)$$

3.5. Extracting Domain Features

By extracting and analyzing the domain characteristics of rumors, the common types and patterns of rumors in this field can be identified more accurately, and more effective detection strategies can be formulated. For example, in the medical and health field, rumors may involve sensitive topics such as disease treatment and drug effects; in the political sphere, rumors may involve sensitive information such as policy interpretations and leaders' remarks. After extracting dual-emotion features, this module combines semantic features with domain features through an attention mechanism to extract more fine-grained features. For the f domain, the domain vector representation is obtained after encoding:

$$D = \{D_1, D_2, D_3 \cdots \cdots D_f\} \quad (19)$$

where D represents the domain eigenvector.

3.6. Prediction Layer

After acquiring the dual-emotion features, they are connected with the word vector embedded in the semantic features extracted by the GRU model and CNN model, and the attention weights are captured with the domain features, which are jointly input into the MLP layer and softmax layer to detect the truth of the rumors:

$$\hat{y} = \text{softmax}\left(\text{MLP}\left(\left[\text{BiGRU}_T, \text{emo}^{\text{dual}}, D\right]\right)\right) \quad (20)$$

4. Experiments and Results

These experiments aimed to verify the validity and practicability of the proposed rumor detection algorithm through a series of carefully designed steps. During the experiments, we selected representative datasets that covered different types of rumors and real information to ensure the comprehensiveness and reliability of the experiment.

In terms of experimental methods, we adopted the BiGRU and CNN to build an efficient rumor detection model through multiple links, such as feature extraction, model training, and parameter tuning. In order to comprehensively evaluate the performance of the model, we used a number of evaluation indicators, such as accuracy, recall rate, F1 score, etc., which can objectively reflect the performance ability of the model in different aspects.

Next, we show the results of the experiments one by one. These results are not only direct proof of the effectiveness of our algorithm but also the basis for subsequent analysis.

4.1. Dataset and Model Parameter Settings

The Weibo-16 dataset has become the benchmark dataset for Chinese rumor detection; each entry is labeled as a rumor or non-rumor [25]. The Weibo-16 dataset contains actual rumor and non-rumor information collected from the Weibo platform. After screening and labeling, these data have high authenticity and can reflect the actual situation of rumors spread on social media. As one of the largest social media platforms in China, Weibo has a wide user base, covering people of different ages, occupations, and regions. Therefore, the Weibo-16 dataset is highly representative in the field of rumor detection.

In the original dataset, the subset of rumors has a higher repetition rate. Considering that redundant data affect the learning and evaluation effect of the trainer, this paper used a text similarity clustering algorithm to eliminate repeated rumors and comments. When splitting the training set and the test set, there is a need to take into account that content duplication can increase the risk of data breaches and make the model tend to learn the characteristics of a particular event (because they may be repeated multiple times during training), which limits the generalization of the model. Therefore, in this experiment, highly similar rumors and comments were filtered out, and a version of Weibo-16 (t) with the elimination of duplicate data was made. Table 2 divides the preprocessed dataset into the training set and test set in a ratio of 3:2.

Table 2. Weibo-16 dataset.

Weibo-16	Training Set		Test Set	
	Text	Comments	Text	Comments
rumor	1410	482,226	941	326,872
non-rumor	806	649,673	554	415,889
total	2211	1,131,899	1495	743,589

The Weibo-16 rumor text contains 1355 rumors and 2351 non-rumors. Among the 1,874,678 comments, real comments account for 43.1%, and fake comments account for 56.9%. The training set is used to train the model classifier, so the number of texts in the training sets is relatively large. After the model is trained, the test set is used to verify the accuracy of the model.

The Chinese-specific affective resource corpus defines the dimension value of each affective feature, and the sub-feature dimensions of the dual affective features are determined from the corresponding affective dictionary, such as d_f , d_e , d_s , and d_a . Based on the existing experimental parameters used by Dr. Zhang Xueyao [25], the relevant data parameters are integrated into the experimental model and are listed in Table 3. d_f is the output of the pre-trained emotion classifier, with a value of 8; d_e is the size of the Chinese emotion category dimension: 21; the emotion score d_s is determined by How Net and has only 1 dimension; d_a represents the number dimension of features: 15; and the full dimension d is obtained by Equation (12) and has a dimension of 66. When extracting comment emotions, the extraction length is set to the first 100 comments; the output dimension of the multi-layer perceptron (MLP) is set to 32 when calculating the accuracy of the rumor.

Table 3. Features and model parameters.

Characteristic Dimension	Parameters
d_f	8
d_e	21
d_s	1
d_a	15
d	66
MLP	32
Extraction length	The first 100 comments

4.2. Evaluation Index

In this paper, precision, recall, and F1 values are used as evaluation indicators to measure the detection performance [30]:

$$P_i = \frac{t_i}{t_i + f_i} \quad (21)$$

$$R_i = \frac{t_i}{t_i + m_i} \quad (22)$$

$$F_i = \frac{2 * R_i * P_i}{P_i + R_i} \quad (23)$$

When classifying the emotion of class text i , t_i represents the number of correctly classified texts, f_i represents the number of incorrectly classified texts, and m_i represents the number of unclassified texts. The higher the F1 score, the better the model performance.

4.3. Experimental Results

4.3.1. Experimental Results of Independent Use of Dual-Emotion Features

In order to verify the validity of dual emotions alone, two baseline features, Emoratio and EmoCred, were selected for comparison, and the same emotion dictionary was used to extract features. Since the semantic features automatically extracted by the rumor detector will cause interference with the verification results, this paper selected a five-layer MLP to exclude interference factors from the rumor detector. The results of MLP detection of six types of emotional characteristics is shown in Figure 7.

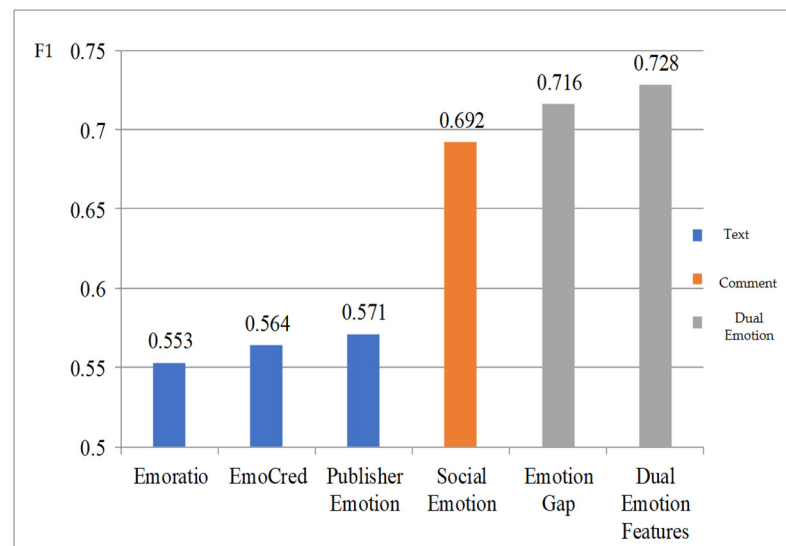


Figure 7. The results of MLP detection of six types of emotional characteristics.

- (1) In the rumor text, the performance of publisher's emotional features is better than the Emoratio and Emocred baseline features;
- (2) The F1 score of social-emotional traits was 12% higher than that of publisher's emotional traits and about 13% higher than that of the two baseline traits;
- (3) The dual-emotion features significantly enhanced the accuracy of the dataset, and the F1 score was as high as 71.6% when only the emotion gap was used.

These data indicate the necessity of dual affective features in affective signal modeling. In addition, the improvement from social affective features and the affective gap in the detection results further proves that the independent use of dual affective features can also improve the effectiveness of rumor detection.

In this paper, five types of emotional features (① emotion category, ② emotion vocabulary, ③ emotion intensity, ④ emotion score, and ⑤ other auxiliary features) are used to extract the emotional features of publishers. To verify the effect of each affective trait, F1 scores were compared before and after one particular type of trait was removed from the dual-emotion features each time. Figure 8 demonstrates that no matter what type of affective feature is removed, the macro F1 score for dual-emotion features decreases. Therefore, it proves the necessity of the existence of the five emotional characteristics, and their combination will achieve the maximum effect.

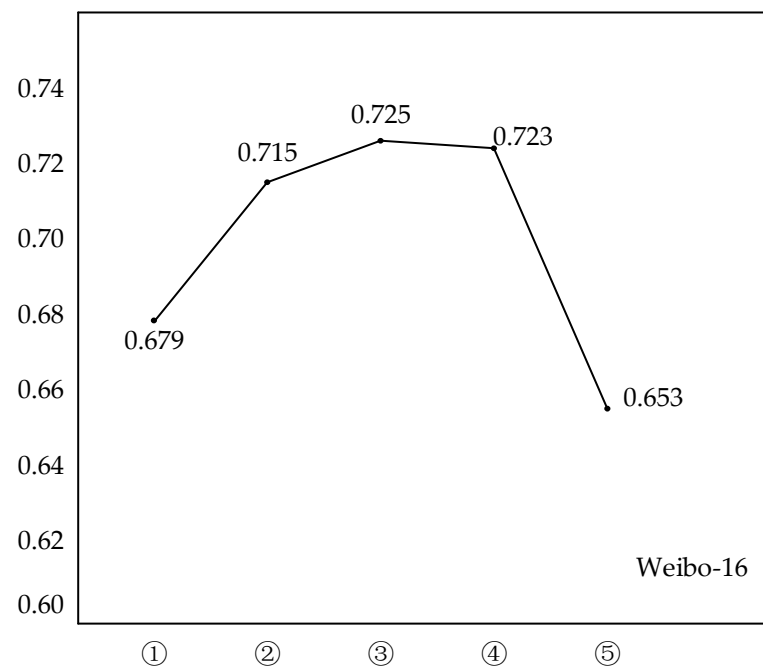


Figure 8. F1 score after subtracting different characteristics from the dual affective characteristics.

4.3.2. Experimental Results of Dual-Emotion Features in Rumor Detector

In the previous section, the validity of dual-emotion features in rumor detection is verified, but because the MLP is a simple multi-layer perceptron without any text features, the representativeness of the experimental results is limited. Next, this paper used different rumor detectors to test the dual-emotion features and analyzed whether dual affective characteristics enhance the effects on rumor detection under the influence of other confounding factors. Then, the two baseline features were introduced into the detector, and the optimization degree of the dual-emotion features was analyzed through data comparison.

In this experiment, the Emoratio and EmoCred baseline features and dual-emotion features were embedded into the BiGRU model and CNN model. Four experimental indicators were obtained: the macroscopic F1 value and accuracy rate of each model, as well as the F1 values of “rumor” and “non-rumor”. The data results in Tables 4 and 5 show that the substitution of dual-emotion features greatly improves the evaluation indexes of the two models. After a detailed analysis, we can see the following:

- (1) In the semantic feature detectors GRU and CNN, the dual-emotion features are added to distinguish between Weibo-16 data categories, and the results show that the accuracy rate of detecting “rumors” and “non-rumors” is increased by 2.7% and 0.7%, respectively.
- (2) In the GRU model, the macro F1 score of dual-emotion features is increased by at least 1.5%, and the addition of Emoratio and EmoCred not only does not improve the performance of the detector but also decreases the index of the macro F1 score. This is because the two baseline features only focus on the rumor text and ignore the emotional information of the comments.
- (3) The addition of bi-affective features improves the performance of the GRU model and CNN model; moreover, comparing the F1 scores of the two detectors combined with those of the Emoratio and EmoCred baseline features reveals that the compatibility and generalization ability of dual-emotion features are more prominent.

Table 4. Results of the BiGRU model.

Model	Macro F1 Score	Precision	F1 Score	
			Rumor	Non-Rumor
BiGRU	0.807	0.822	0.754	0.860
+Emoratio	0.794	0.810	0.738	0.851
+EmoCred	0.766	0.778	0.711	0.820
+Dual-emotion features	0.826	0.838	0.781	0.871

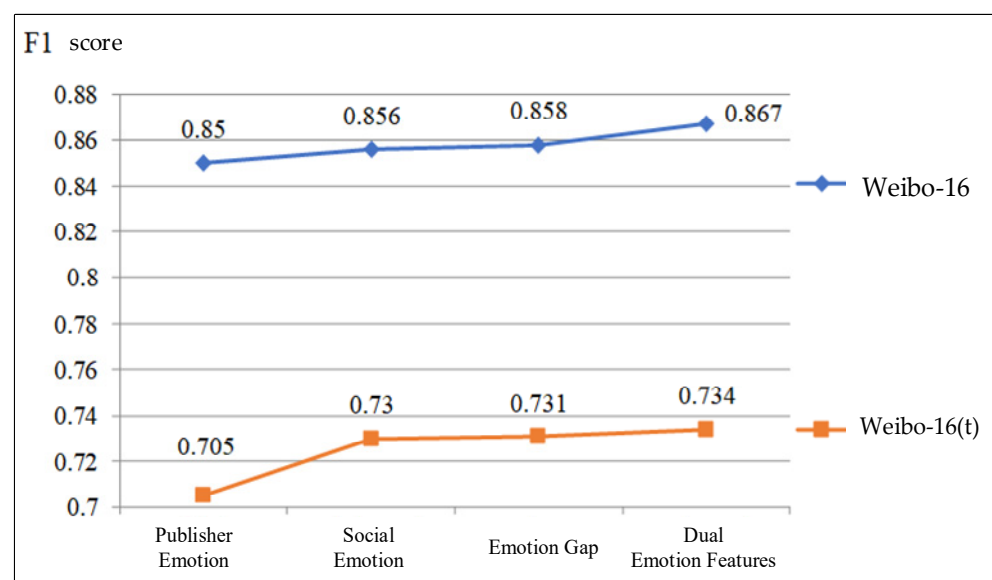
Table 5. Results of the CNN model.

Model	Macro F1 Score	Precision	F1 Score	
			Rumor	Non-Rumor
CNN	0.824	0.845	0.762	0.886
+Emoratio	0.837	0.857	0.780	0.894
+EmoCred	0.849	0.867	0.797	0.901
+Dual-emotion features	0.871	0.883	0.832	0.911

4.3.3. Experimental Results of Emotional Characteristics in the Ablation Experiment

In this experiment, the data in Weibo-16 were selected to be re-split according to a time series. The latest 20% of the data were selected as the test set. Of the remaining 80% of the new dataset, the latest 25% was selected for validation, and the rest of the data were used as the training set. In this way, model training can avoid over-fitting and adapt to the update and development of network rumors to the greatest extent.

As can be seen in Figure 9, the F1 value reached 86.7% after adding dual-emotion features to the CNN detector, and the enhancement effect of social emotions and the emotion gap was significantly better than that of publisher's emotional features. As shown in Figure 10, although the fluctuation in the data is relatively smooth, the detection performance of dual-emotion features is still the best, and the enhancement effect of publisher's emotional features on rumor detection is low.

**Figure 9.** Detection results of CNN model on Weibo-16 and Weibo-16 (t).

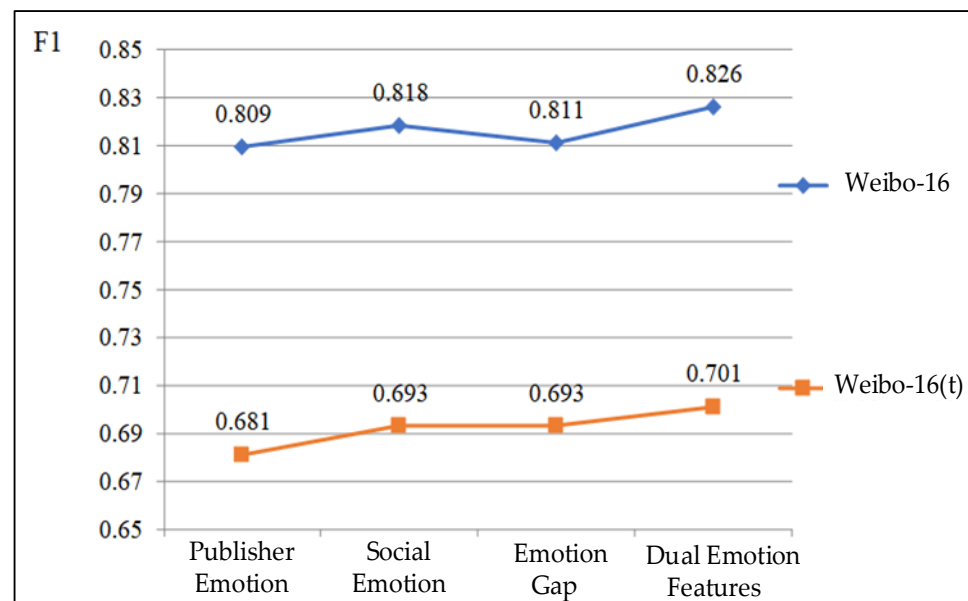


Figure 10. Detection results of BiGRU model on Weibo-16 and Weibo-16 (t).

The results of these two models on the two datasets, Weibo-16 and Weibo-16 (t), were comprehensively analyzed. The three components of the dual-emotion features can be used independently in any model to enhance the effect; the improvement of the macro F1 score by supplementing the data with social-emotional features or the emotion gap is greater than that of the publisher's emotional features, and the performance of social emotions and the emotion gap is relatively consistent. Therefore, among the three emotional characteristics, social emotion and the emotion gap can play a greater role in detection. It is found that, compared with social comments, there are fewer emotional resources available to extract the emotions of publishers, which leads to the insufficient analysis of texts. However, the comment base of rumors is large, which provides sufficient emotional resources for the extraction of emotional features.

4.3.4. Comparative Results of Relevant Studies

In this paper, the BiGRU and CNN models were used with dual-emotion features, and the data show that dual emotions can better enhance the performance of the rumor detector. However, the results of the two models alone cannot comprehensively summarize the characteristics of dual emotions. Therefore, this section will refer to the previous experimental results on the dataset of Weibo as a supplement. The detection data for DTRank, SVMRBF, and DTC initial models are compared with the results of the two models introduced in this paper. DTRank, SVM-RBF, DTC, and BiGRU+ models have their own advantages and characteristics in the field of rumor detection. Choosing these models as baseline models enables a comprehensive evaluation of the performance of the newly proposed rumor detection models and promotes the continuous development and innovation of rumor detection technology.

Table 6 evaluates the effectiveness of the detectors from four aspects: accuracy, precision, recall and F1 value.

Table 6. Experimental results of different models on Weibo-16.

Model	Tag	Accuracy	Precision	Recall	F1
DTRank	F	0.732	0.738	0.715	0.726
	T		0.726	0.749	0.737
SVMRBF	F	0.818	0.822	0.812	0.817
	T		0.815	0.824	0.819
DTC	F	0.831	0.847	0.815	0.831
	T		0.815	0.847	0.830
BiGRU+	F	0.879	0.830	0.839	0.835
	T		0.907	0.902	0.905
CNN+	F	0.883	0.873	0.795	0.832
	T		0.889	0.934	0.911

- (1) After adding bi-affective features, the accuracy of BiGRU and CNN models on the Weibo-16 dataset reached 87.9% and 88.3%, which increased the accuracy by at least 4.8–5.2 percentage points compared with the DTC model with good detection performance;
- (2) In the classification of “rumor” and “non-rumor”, the accuracy, recall rate, and F1 value were significantly improved, and the various indicators reached more than 90%.

The above data analysis shows that dual-emotion features play a prominent role in rumor detection, which can enable the model to obtain higher accuracy. The above data are sufficient to show that bi-affective features serve an important auxiliary function in the rumor detection task.

5. Conclusions and Future Work

In this paper, we have established a model of dual-emotion signals for rumor detection, leveraging the unique characteristics derived from the similarity and difference between publisher emotions, social emotions, and dual emotions. Recognizing the limitations of single-emotion feature analysis, we have introduced semantic-feature-based BiGRU and CNN models for fine-grained text analysis. Through comparative experimental analysis on multiple datasets, we have demonstrated the feasibility and progress of our proposed models. The experimental results show that the accuracy of the BiGRU and CNN models on the Weibo-16 dataset reaches 87.9% and 88.3%, respectively, surpassing the DTC model with good detection performance by at least 4.8–5.2 percentage points. In particular, for the “rumor” and “non-rumor” categories, the accuracy, recall, and F1 values have been significantly improved, with each indicator exceeding 90%.

However, it is important to acknowledge that this model still faces some limitations:

- (1) **Data dependency:** Our methods are highly dependent on the quality and integrity of the collected data. When the data are sparse, noisy, or biased, the accuracy and generalization ability of the model may be affected to some extent. This limitation underscores the need for careful data preprocessing and validation. In future studies, we will further explore how to effectively process imperfect data to improve the robustness of the model, ensuring its reliability even in challenging data environments.
- (2) **Computational complexity:** For large datasets, our approach may require substantial computational resources and time costs. This can pose challenges in real-world applications, particularly those requiring real-time analysis. To address this limitation, we plan to focus on optimizing the algorithm design and improving computational efficiency in future work. By refining our models and leveraging advanced computational techniques, we aim to make our approach more scalable and suitable for large-scale data processing and real-time analysis scenarios.

In the next phase of our research, we will endeavor to extract emotional features from a broader range of sources, including images, text embedded in images, animations (GIFs), and videos. Furthermore, we will continue to enhance the classifier's ability to detect the latest data and real-time hot topics, thereby expanding the applicability and effectiveness of our rumor detection model.

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